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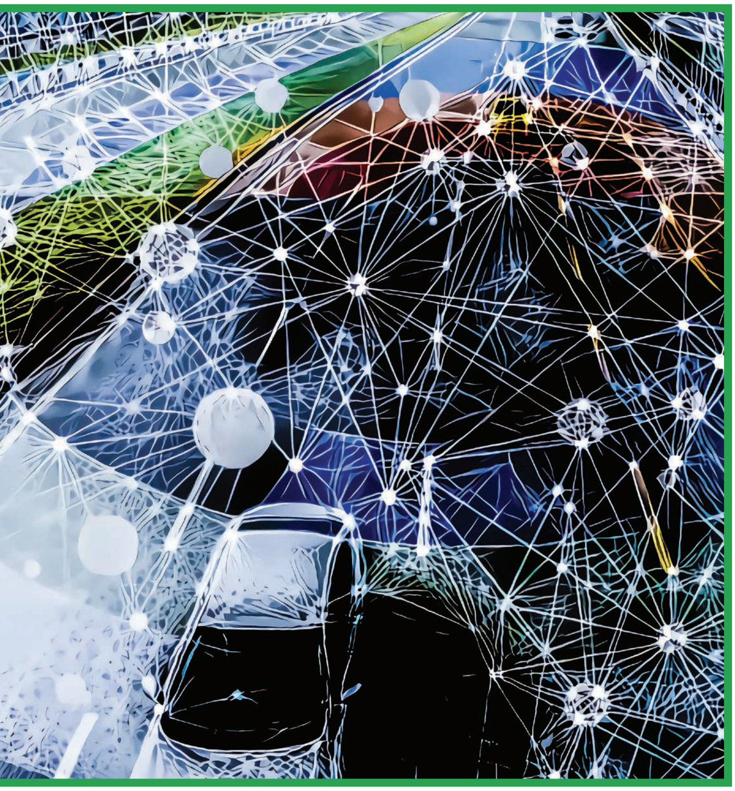
Radar Interference Mitigation for Automated Driving

Exploring proactive strategies



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Autonomous driving relies on a variety of sensors, especially radars, which have unique robustness under heavy rain/fog/snow and poor light conditions. With the rapid increase of the amount of radars used on modern vehicles, where most radars operate in the same frequency band, the risk of radar interference becomes a compelling issue. This article analyzes automotive radar interference and proposes several new approaches that combine industrial and academic expertise toward the goal of achieving interference-free autonomous driving (AD).

Introduction and motivation

Radar is becoming standard equipment in all modern cars, supporting, e.g., cruise control and collision avoidance in most weather conditions while providing high-resolution detections on the order of centimeters in the millimeter-wave (mm-wave) band. The next generation of advanced driver assistance (ADAS) and AD vehicles will have a multitude of radars covering numerous safety and comfort applications such as crash avoidance, self-parking, in-cabin monitoring, cooperative driving, collective situational awareness, and so on. Because automotive radar transmissions are uncoordinated, there is a nonnegligible probability of interference among vehicles, as shown in Figure 1. Although current automotive radars have already been impacted by interference to some extent, today, they are unlikely to raise customer awareness because state-of-the-art automotive radars are continuously updated and improved upon at many system levels.

However, the mutual interference problem is expected to become more challenging if not properly handled, as more vehicles are equipped with a greater number of radars providing 360° situational awareness at various distances that enable more advanced future ADAS and AD functionalities. This is evidenced by multiple international studies, such as the MOSARIM project [1] and the more recent IMIKO radar project. All of the major players in the automotive sensor market, such as Volvo and Veoneer, are studying the next generation of "interference-free radars." This includes, for example,

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enhancing models to determine the impact of a larger density of radars, simulating new interference scenarios, and investigating different medium access control (MAC) models and methods to coordinate radar transceivers, both decentralized and centralized.

At this point, the automotive industry is ready to consider novel designs and approaches, which may impact standardization bodies before a new frequency spectrum is made available in the higher radio-frequency (RF) bands. Signal processing can provide ways to reduce or mitigate interference, both at the raw signal level as well as at the postdetection/target-tracking level. The particular properties and requirements of automotive radar impose significant challenges in terms of signal processing. This includes the combination of radar and communication waveforms, which brings up further possibilities regarding ultrareliable low-latency communications in vehicular ad hoc networks (VANETs). It is therefore important and timely to review what has been done, what the reasonable approaches are, and what the future holds.

The focus of this article is on frequency-modulated continuous wave (FMCW) radar because it is the most common and robust automotive radar. We provide an analysis of the impact of interference in FMCW both quantitatively and qualitatively, in terms of their probability, severity, and effects. Then, we cover different ways to mitigate interference, ranging from changing FMCW parameters to new signal structures and the explicit coordination between vehicles. We also study new techniques that are potentially more robust toward interference, including stepped-frequency orthogonal frequency-division multiplexing (OFDM). Finally, we describe what we believe will be the long-term evolution of automotive radar and its relation to mobile communication.

Automotive radar

History and future of automotive radar

Radar has been used in automotive applications for a long time. In 1949, unfortunate car drivers were issued speeding tickets based on speed measurements obtained from the radar speed gun, recently invented by John L. Barker. However, on-board automotive radar was not made commercially available until 1999, when it was introduced for collision warning and automatic cruise control (ACC). See [2] for an early history of automotive radar with some entertaining vintage photography. Over the years, there has been a strong push to increase the integration level of mm-wave electronics used for automotive radar and industrial radar sensors. The early discrete hardware designs have been replaced by a few chips in III–V-materials, and now, CMOS single-chip solutions are available. CMOS technology provides the ability to fully integrate analog and digital electronics, making very advanced protocols and detection schemes possible at low cost and low power. Consequently, radar is becoming more and more common for supporting various automotive applications. ADAS systems based on radar are today standard equipment in most new vehicles. Vehicles capable of some level of AD are also expected to rely, at least to some degree, on radar systems for monitoring vehicle surroundings. The number of radar transceivers operating throughout the traffic environment is foreseen to increase rapidly over the coming years. As the number of radar transceivers in the traffic infrastructure increases, radar interference is also expected to increase.

Today, most radar transmissions are uncoordinated, meaning that there is no a priori agreement on who is allowed to transmit and when. A number of recent studies have identified the interference situations that are likely to arise as the automotive radar transceiver market penetration increases [1], [3]. FMCW waveforms can, up to a point, be relatively easily repaired in the event they are intermittently corrupted by interference [4], which is why they are still operational. Future radar systems are expected to occupy frequency bands that are higher and higher in frequency. Transceivers operating around 77 GHz are available today, and transceivers operating at carrier frequencies beyond 100 GHz are expected. Frequencies as high as 300 GHz and beyond are being considered for some applications, such as synthetic aperture radar mapping. Operation at such high frequencies brings the obvious benefits of improved miniaturization, but also presents challenges in terms of hardware complexity and signal attenuation. Moreover, interference-free operation will require radar transmission standardization. A standardized transmission scheduling system resembling today's cellular communication system

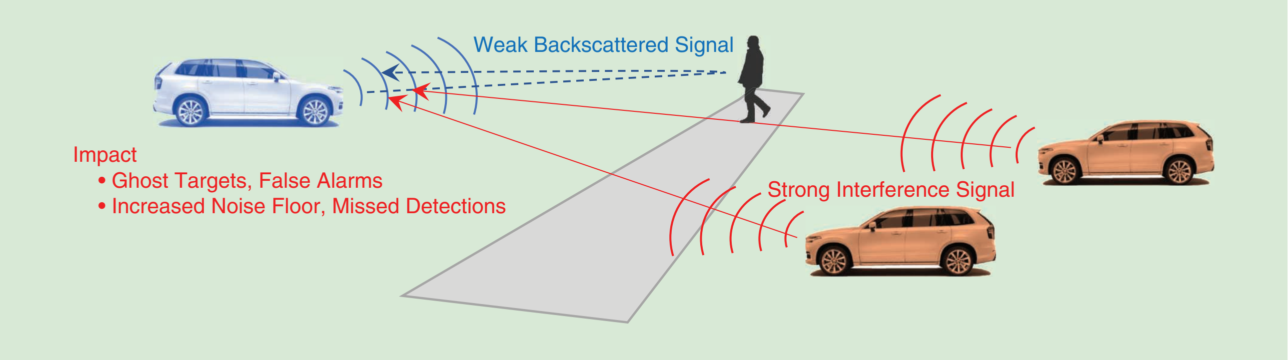

FIGURE 1. Interference is generally much stronger than the desired radar signal, due to its one-way propagation. Interference increases with more interfering radars and leads to false alarms and missed detections.

would present a solution to the interference problem, but it is not without challenges, both technical and political.

Basics of FMCW radar

In a general FMCW radar, a frequency sweep, i.e., a chirp, is generated by a voltage-controlled oscillator operated by a digital synthesizer. The generated chirp signal is split and sent into two different signal paths: one path is directed to the transmitter (Tx) antenna, while the other path is directed to the mixer correlator. Before the chirp is sent out on the Tx antenna it passes a power amplifier (PA), which boosts the transmitted energy. Here we assume that there are no idle periods between consecutive chirps. It is also possible to introduce random interchirp idle periods to reduce the probability of mutual interference between FMCW radars. The transmit waveform of an FMCW radar with K consecutive linear FM chirps (or sweeps) can be expressed as [5]

$$s(t) = \sum_{k=0}^{K-1} x(t - kT), \tag{1}$$

where the individual chirps are given by

$$x(t) = e^{j\varphi(t)}\mathrm{rect}_T(t), \quad \varphi(t) = 2\pi(f_c t + 0.5\alpha t^2). \tag{2}$$

Here, $\alpha = B/T$ is the chirp slope, B denotes the sweep bandwidth, T represents the chirp duration, f_c is the carrier frequency, and $\mathrm{rect}_T(t)$ is square pulse of duration T with an amplitude of 1. The received reflected signal from a target is very weak due to its two-way free-space propagation path loss and the losses incurred during reflection and thus needs to be amplified with a low-noise amplifier (LNA) to maintain an acceptable signal-to-noise-ratio (SNR). The amplified signal from the target reflection is correlated with the Tx signal in the mixer correlator, which is called *dechirping*. The low-pass filter at the output of the correlating mixer offers some interference rejection. The round-trip delay and Doppler shift caused by the relative velocity of the target shifts the frequency of the received signal compared to that of the transmitted signal. As a result, the mixer creates a beat signal that will pass through a low-pass filter and be digitized, yielding delay and Doppler estimates after matched filtering. In modern automotive radars, it is also possible to estimate azimuth and elevation of targets using multiple antennas.

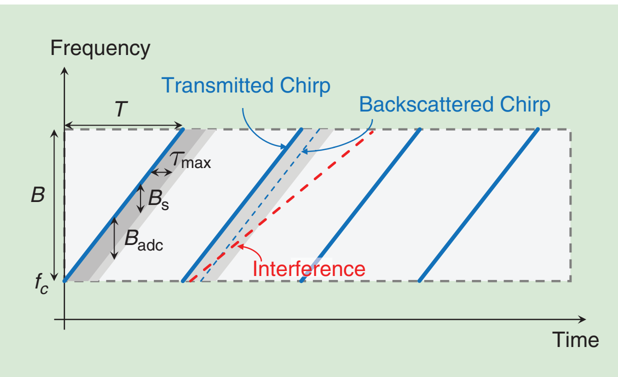

FIGURE 2. Four consecutive chirps in time-frequency representation, with several key notations included. Backscattered and interfering signals are shown at the second chirp.

FMCW radar signal processing chain

Suppose there exists a single target of interest acting as a point scatterer, characterized by a complex channel gain γ [including the effects of path loss, antenna gain, and radar cross section (RCS)], an (initial) round-trip propagation delay $\tau = 2R/c$, and a normalized relative Doppler shift $\nu = 2v/c$, where R and v denote, respectively, the distance and relative radial velocity between the radar and target, and c is the speed of wave propagation. The received backscattered signal is now processed in three stages.

Stage 1: Dechirping

Under the stop-and-hop assumption [6, Ch. 2.6.2], the kth chirp of the received signal is given by

$$r_k(t) = \gamma x(t + (t + kT)\nu - \tau) + w_k(t), \tag{3}$$

where $0 \le t \le T$ denotes the time relative to the beginning of the kth chirp, and $w_k(t)$ is measurement noise. To obtain the beat signal at the intermediate frequency, the received signal $r_k(t)$ in (3) is dechirped through conjugate mixing with the transmitted signal $x(t)$ in (2). Here we ignore the terms whose total phase progression over a coherent processing interval (CPI) of K chirps is smaller than $\pi/4$ for typical automotive FMCW settings [6, Ch. 2.6.3]:

$$y_k(t) = r_k(t)x^*(t) = \gamma e^{j2\pi f_c \nu(t+kT)} e^{-j2\pi\alpha\tau t}\mathrm{rect}_T(t-\tau) + w_k(t)x^*(t). \tag{4}$$

Let $\tau_{\max}$ denote the round-trip delay (see Figure 2) that corresponds to a maximum target range of interest (i.e., $\tau_{\max} \ge \tau$), which is related to the radar bandwidth of interest B_s as $\tau_{\max} = B_s/\alpha$. The analog-to-digital converter (ADC) bandwidth $B_{adc} \ge B_s$ imposes a limit on B_s and thus the maximum detectable range $\tau_{\max}$. After low-pass filtering the beat signal in (4) with bandwidth B_s, sampling with a period of T_s for $\tau_{\max} \le t \le T$, we rearrange into a slow-time fast-time data matrix, where the kth row contains the samples of the kth chirp (fast time), while the nth column contains the nth sample of each chirp (slow time). In other words, we have

$$y_{k,n} = \gamma e^{j2\pi(-\alpha\tau + f_c\nu)nT_s} e^{j2\pi f_c \nu kT} + w_{k,n}, \tag{5}$$

for $k = 0, \ldots, K-1$ and $n = n_{\max}, \ldots, N-1$, $n_{\max} = \lfloor \tau_{\max}/T_s \rfloor$, $N = \lfloor T/T_s \rfloor + 1$, and $w_{k,n}$ are independent and identically distributed complex Gaussian noise samples with variance σ^2.

Stage 2: Target range-velocity estimation

To provide an estimate of target range and velocity, a 2D discrete Fourier transform (DFT) can be applied across slow- and fast-time dimensions of the beat signal in (5), which yields the

FMCW delay-Doppler spectrum evaluated at a given delay-Doppler pair $(\hat{\tau}, \hat{\nu})$:

$$z(\hat{\tau}, \hat{\nu}) = \sum_{k=0}^{K-1} \sum_{n=n_{\max}}^{N-1} y_{k,n} e^{j2\pi\alpha\hat{\tau}nT_s} e^{-j2\pi f_c \hat{\nu} kT}. \tag{6}$$

The periodogram $|z(\hat{\tau}, \hat{\nu})|^2$ corresponding to (6) yields a dominant target peak at $(\hat{\tau}, \hat{\nu}) = (\tau - f_c \nu/\alpha, \nu)$, which can be recovered using, for example, constant false alarm rate (CFAR) detectors [6, Ch. 6]. The peak value of $|z(\hat{\tau}, \hat{\nu})|$ is proportional to the processing gain $G_{\mathrm{p}} = K(N - n_{\max})$. This frequency identification method is referred to as the *periodogram spectral estimator* [7, Ch. 2.2.1]. Here, the shift $f_c \nu/\alpha$ in delay stems from range-Doppler coupling inherent in the FMCW waveform [6, Ch. 4.6.4]. To compensate for the coupling effect, the Doppler-dependent term $f_c \nu/\alpha$ can be added back to the delay estimate after delay-Doppler retrieval. When there are multiple objects in the radar field of view (FOV), (6) will have multiple peaks. To distinguish the different objects, each object pair must be separated by a certain minimum gap in delay and Doppler domains, which is determined by the radar resolution. The range and velocity resolution of an FMCW radar can be derived from (6) (assuming $n_{\max} \ll N$) as $\Delta R = c/(2B)$ and $\Delta v = \lambda_c/(2KT)$, where λ_c is the carrier wavelength [7, Ch. 2.4]. Therefore, higher sweep bandwidth leads to better range resolution, while a longer CPI duration means improved velocity resolution.

Stage 3: Tracking filter

At the final stage of the signal processing chain, the delay-Doppler detections $\{(\tau_p, \nu_p)\}_{p=0}^{P-1}$ (along with the corresponding azimuth-elevation pairs in the case of multiple antennas [5]) are fed to a data association and tracking filter to provide filtered 3D positions and the velocities of surrounding objects. Here, P denotes the number of targets seen during one scan.

Is interference really a problem?

In the next section, we provide a theoretical analysis for the impact of interference on the radar signal processing. We start by studying a single link and then extend to a network of vehicles on a multilane highway to assess the impact of interference as a function of the vehicle and radar density as well as the deployment scenario. Our focus will be on direct interference from one radar to another. Indirect interference (i.e., scattered on objects) will be weaker and is ignored for conceptual simplicity.

Single-link interference

The interfering radar employs the FMCW waveform $s_{\mathrm{int}}(t) = \sum_{k=0}^{K-1} x_{\mathrm{int}}(t - k\tilde{T})$ where $x_{\mathrm{int}}(t) = e^{j2\pi(f_c t + 0.5\tilde{\alpha}t^2)} \mathrm{rect}_{\tilde{T}}(t)$, while the victim radar utilizes the same waveform as specified in (1) and (2). Here, $\tilde{\alpha} = \tilde{B}/\tilde{T}$, $\tilde{B}$ and $\tilde{T}$ denote, the chirp slope, sweep bandwidth, and chirp duration of the interfering radar, respectively. The samples (5) then become

$$y_{k,n} = \gamma e^{j2\pi(-\alpha\tau + f_c\nu)nT_s} e^{j2\pi f_c \nu kT} + \gamma_{\mathrm{int}} x_{\mathrm{int},k,n} + w_{k,n}.$$

Interference is generally much stronger than the desired backscattered signal because they are governed by the Friis free-space propagation equation and the radar equation, respectively [8]:

$$|\gamma_{\mathrm{int}}|^2 = P\frac{G_{\mathrm{trx}}\lambda^2}{(4\pi)^2 r^2}, \quad |\gamma|^2 = P\frac{G_{\mathrm{trx}}\sigma\lambda^2}{(4\pi)^3 d^4}, \tag{7}$$

where r is the distance between the interferer and the victim radar, d is the distance between the radar and the target, P is the transmit power, G_{trx} is the combined transmit and receive antenna gain, and σ is the RCS of the target. Hence, for similar d and r and typical values of σ, $|\gamma_{\mathrm{int}}|^2 \gg |\gamma|^2$. The nature of the interference depends on the total interference power (i.e., the aggregate power of the interference samples) and the level of coherence between the victim and interfering radar [9].

The total interference power, which is a function of the radar waveform parameters and signal delays, depends on the statistics of the samples $x_{\mathrm{int},k,n}$. The samples satisfy $|x_{\mathrm{int},k,n}|^2 \in \{0, 1\}$ depending on whether or not the interference signal at time (k, n) is in the bandwidth of interest of the victim radar. As a result, the overall power of the interference is $\mathbb{E}\{\sum_{k,n} |\gamma_{\mathrm{int}}|^2 |x_{\mathrm{int},k,n}|^2\} = f|\gamma_{\mathrm{int}}|^2 G_{\mathrm{p}}$, where G_{p} is the radar processing gain and f is the average interference probability.

How this total interference power manifests itself depends on its radar parameters, and interference can be classified as *coherent*, *incoherent*, or *partially coherent* [10]. Coherent interference occurs when the interferer uses the same parameters (α, T, and B) as the victim radar. In that case, the interfering radar signal leaking into the bandwidth of interest B_{s} of the victim radar (i.e., $f = 1$) leads to a ghost target, i.e., a peak in the delay-Doppler spectrum with very high power [11]. Ghost targets lead to false detections, which in turn may cause incorrect behavior of safety systems. Incoherent interference occurs when samples $x_{\mathrm{int},k,n}$ are independent random variables due to the interferer using very different waveform parameters (e.g., a different chirp pattern) so that the total interference power $f|\gamma_{\mathrm{int}}|^2 G_{\mathrm{p}}$ ends up as an increased noise floor. A noise floor increase resulting from interference can lead to more severe degradation in detection performance than an equivalent increase in thermal noise floor due to the sidelobes of the interference spectrum [10]. In between these two extreme cases, partially coherent interference occurs due to a slight mismatch in chirp slope or chirp duration or in the presence of phase noise (PN). This causes the energy of the ghost target peak (caused by coherent interference) to spread over the delay-Doppler domain. For incoherent or partially coherent interference, where the interference manifests itself as a noise floor, the overall interference power [or, equivalently, the average signal-to-interference-plus-noise ratio (SINR)] is a reasonable performance metric, while for fully coherent interference, percentiles are more meaningful.

To illustrate how a ghost target is spread out depending on the relative waveform parameters, Figure 3 shows the fast-time fast Fourier transform (FFT) output, i.e., the range FFT, corresponding to an interfering radar signal as a function of distance. The larger the difference in the chirp slope, the more the interference is spread out, due to the decrease in coherence; this affects the detection of targets in various ways. Incoherent interference may hinder the detection of low RCS targets (such as pedestrians and cyclists) over a large fraction of the delay-Doppler domain, whereas a (partially) coherent interference can mask even high RCS targets (vehicles) but in a smaller fraction of the delay-Doppler domain.

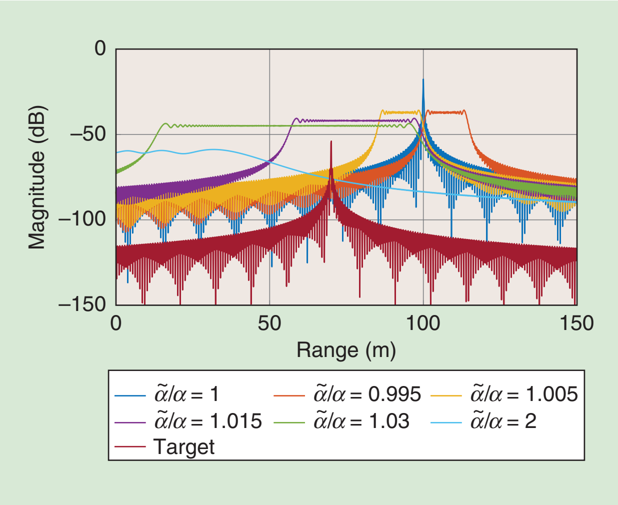

FIGURE 3. The FMCW range profiles in the presence of an interfering radar for various values of chirp-slope ratios, where $\tilde{\alpha}$ and α represent the chirp slopes of the interfering radar and the victim radar, respectively. The victim radar waveform parameters are $f_c = 77$ GHz, $B = 1$ GHz, and $T = 20\ \mu$s. The interfering radar has an identical chirp duration of $\tilde{T} = 20\ \mu$s but a varying sweep bandwidth of $\tilde{B}$ (thus, $\tilde{\alpha}$). The interference signal has a one-way propagation delay τ_{int} corresponding to a range of $R_{\text{int}} = 100$ m, while the desired target is located at $R = 70$ m. Due to an increased noise floor, the target may not be detected depending on its range and the chirp-slope mismatch between the victim and interfering radars.

In practice, oscillators in FMCW radars do not have an ideal, impulse-like RF spectrum due to phase and frequency instabilities [12], [13]. In Figure 4, we demonstrate the effect of oscillator PN on the averaged range response of a victim FMCW radar when the oscillators of both the victim and interfering radars are subjected to phase noise processes with parameters $L_p = -70$ dBc/Hz (pedestal height) at $W_p = 200$ kHz (pedestal width) [13]. Range spectra are derived by computing the range FFTs of signal and interference powers averaged over the randomness of PN which is modeled as a zero-mean, wide-sense stationary random process under the assumption of white FM PN in the oscillator [14, Sec. V]. As observed from Figure 4, the oscillator PN induces the spectral smearing of target and interference profiles, thereby causing a loss of details in the spectrum, which deteriorates detection performance and leads to the masking of weak targets.

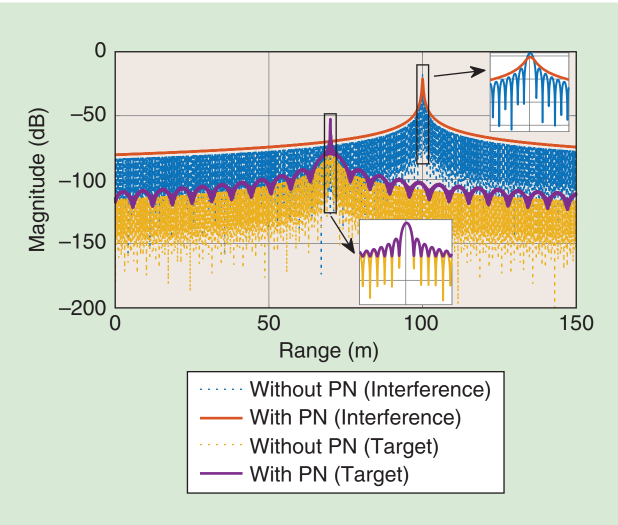

FIGURE 4. The range profiles of a victim FMCW radar in the presence of an interfering radar with identical chirp parameters (coherent interference), where both radars' oscillators have PN processes (the profiles without PN are also shown for comparison). The same parameters as those that are shown in Figure 3 are used with $\tilde{\alpha}/\alpha = 1$. The perfect range decorrelation of the interfering signal with the victim radar signal (due to independent PN processes at the victim and interfering radars) makes the spectral smearing effect more pronounced in the interference profile than in the target profile [15].

Network interference

The aforementioned interference analysis can be extended to a complete network, for instance, on a multilane road, by employing a stochastic geometry approach [8]. As shown in Figure 5, consider a victim radar surrounded by L lanes of traffic, with lane separation R, each modeled as a 1D Poisson point process $\Phi(\mathbf{x})$ with intensity $1/\Delta$ (so that Δ is the expected distance between vehicles and $\mathbf{x}$ is a vehicle location along a road) [3]. Radars (here, one per side of the vehicle) are incoherent and can have different chirp durations, but otherwise share the same bandwidth B, duty cycle $u \in [0, 1]$ (i.e., the fraction of time the radar is transmitting), and FOV.

We recall that for antennas with a narrow FOV, the antenna gain is $G \approx 4\pi/(\phi\theta)$, where ϕ is the beamwidth in the elevation domain and θ the beamwidth in the azimuth domain. This means that a radar with 1 and 30° elevation and azimuth beamwidth, respectively, will have a gain of approximately 31.3 dBi. The expected value of the interference probability f is easily found to be $f = u\alpha\tau_{\max}/B$. Here we made use of the following asymptotic results: 1) when $\tilde{\alpha} = \alpha$, the probability of interference is uB_s/B and the interference lasts an entire chirp duration; 2) when $\tilde{\alpha} \ll \alpha$, the probability of interference is u and the duration is $\tilde{\alpha}\tau_{\max}/(\alpha - \tilde{\alpha})$;

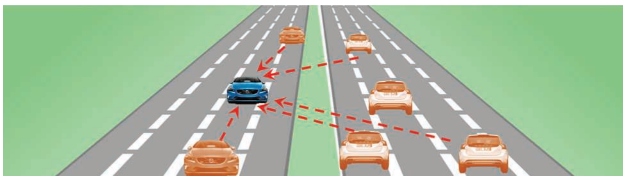

FIGURE 5. Network interference on a six-lane highway. Interference from interfering radars is aggregated and depends on the properties of the individual radars as well as on the placement of vehicles on the road.

and 3) when $\tilde{\alpha} \gg \alpha$, there are $\tilde{\alpha}/\alpha$ simultaneous interferers, each lasting for a duration of $\alpha\tau_{\max}/(\tilde{\alpha}-\alpha)$. As a result, the aggregate interference seen by the victim radar due to interference from lane $\ell \in \mathbb{Z}$ (indexed with reference to the victim radar) is

$$I_{\mathrm{p}}(\ell) = \sum_{\mathrm{x} \in \Phi(\mathrm{x}) \cap \mathrm{x} \in \mathrm{FOV}} Pf \frac{G_{\mathrm{trx}}\lambda^2}{(4\pi)^2 r^2(\mathbf{x})}, \tag{8}$$

where $r(\mathbf{x}) = \sqrt{\ell^2 R^2 + x^2}$, where x is the 1D position along the road ranging from $\ell R/(\tan\theta_{\mathrm{p}}/2)$ to $+\infty$. Here, θ_{p} is the minimum of the forward and backward FOV. Accordingly (with a slight abuse of notation), the interference averaged over the locations of the interferers is

$$\begin{aligned}\mathbb{E}\{I_{\mathrm{p}}(\ell)\} &= P\frac{G_{\mathrm{trx}}\lambda^2}{(4\pi)^2}\frac{f}{\Delta}\int_{\frac{\ell R}{\tan\theta_{\mathrm{p}}/2}}^{+\infty}\frac{1}{\ell^2R^2+x^2}\mathrm{d}x \\ &= P\frac{G_{\mathrm{trx}}\lambda^2}{(4\pi)^2}\frac{f}{\Delta}\frac{1}{\ell R}\frac{\theta_{\mathrm{p}}}{2},\end{aligned} \tag{9}$$

while for $\ell = 0$, $r(\mathbf{x}) = x$, where we need a certain safety margin to avoid singularities, we set x from Δ to $+\infty$, leading to $I_{\mathrm{p}}(\ell=0) = P(G_{\mathrm{trx}}\lambda^2/(4\pi)^2)f(1/\Delta^2)$. An example of network interference on a six-lane highway is shown in Figure 6, as a function of the average intervehicle spacing Δ for a vehicle target 150-m away with an RCS of 10 m^2. The analytical result shows the impact of the interference of nearby vehicles, leading to an orders of magnitude reduction of the SINR. For small Δ, the interference is larger from passing lanes, while for large Δ, oncoming traffic dominates. We also observe that, even though the interference power can be large, factor f reduces its impact significantly. The example $f \approx 0.01$ leads to a 20-dB reduction in interference. In this example, the target can be detected in spite of incoherent network interference, while a pedestrian target with a smaller RCS (such as 0.1–1 m^2) farther away than 50 m would be difficult to detect.

Intermediate conclusion

From the aforementioned analysis, we found that interference can manifest itself in different ways and increase the occurrence of both missed detections and false alarms. Due to the nature of FMCW signals (in particular, the small instantaneous bandwidth), there is a natural robustness to interference. Both the total received interference power and the mutual coherence between victim and interfering radar play an important role. As a rule of thumb, the signal-to-interference ratio (SIR) for a target at distance d due to the power transferred by an interferer at distance r to a victim radar can be determined as follows: the useful signal power (the peak of the periodogram) and the interference power are

$$S = |\gamma|^2 G_{\mathrm{p}}^2, \quad I \leq f|\gamma_{\mathrm{int}}|^2 G_{\mathrm{p}} G_{\mathrm{I}}, \tag{10}$$

where $G_{\mathrm{I}} \in [1, G_{\mathrm{p}}]$ depending on the level of coherence of the interference (i.e., $G_{\mathrm{I}} = 1$ for incoherent interference and $G_{\mathrm{I}} = G_{\mathrm{p}}$ for coherent interference). The level of coherence can be characterized through signal-to-interference mitigation gain, which is a function of FMCW waveform parameters of victim and interfering radars [10], [16]. Hence,

$$\mathrm{SIR} \geq \frac{|\gamma|^2 G_{\mathrm{p}}^2}{f|\gamma_{\mathrm{int}}|^2 G_{\mathrm{p}} G_{\mathrm{I}}} = \frac{\sigma r^2}{(4\pi)d^4}\frac{G_{\mathrm{p}}B}{u\alpha B_{\mathrm{s}}G_{\mathrm{I}}}. \tag{11}$$

The first factor is out of the designer's control, while the second factor can be optimized via the duty cycle (small u), chirp slope (small B_{s}/B), FOV (thereby reducing f), and effective processing gain (increase $G_{\mathrm{p}}/G_{\mathrm{I}}$) to ensure that the SIR is much large than 1. Our results indicate that incoherent interference leads to a significant increase in the noise floor (tens of dBs) so that it can reduce the ability to detect weak targets. Nevertheless, for nearby targets or targets with a high RCS, the SIR margin is sufficient to allow for reliable detection. When the interference is partially coherent this margin drops significantly.

Interference mitigation strategies

The impact of interference includes ghost targets and increases in noise floor. Both are detrimental to radar operations. Approaches that deal with interference can be grouped as either reactive, which aim to reduce the impact of interference after it has occurred, or proactive, which aim to avoid or reduce interference by design. In the next section, we describe various reactive strategies as well as three such proactive strategies: a quasi-orthogonal FMCW waveform, a low-rate data communication between radar Txs, and an OFDM radar approach. Each of these approaches lead to high interference suppression, with limited impact on radar performance. In addition, the OFDM

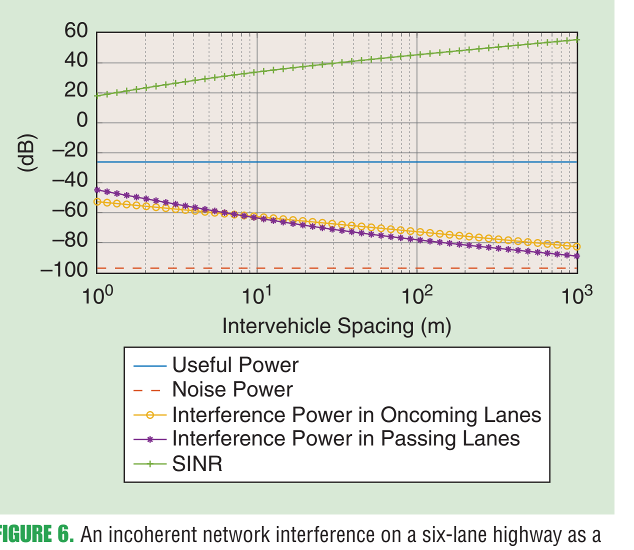

FIGURE 6. An incoherent network interference on a six-lane highway as a function of average vehicle spacing. The distance to the target is 150 m, with a 10-μs chirp time, $\sigma = 10\,\mathrm{m}^2$, and 10 dBm of transmit power. It features a 1-GHz bandwidth, $T = 30\ \mu$s, a 150-m maximum range, a 50-MHz ADC bandwidth, 10 dBm of transmit power (the same for the front and back ends), a 20% duty cycle, a 100-μs frame duration, and 30 and 90° FOVs in the forward and backward directions, respectively.

radar approach can enable high-rate communication links between vehicles.

Standard (reactive) approaches

Extensive studies have been conducted in the context of the MOSARIM project, where a broad range of time- or frequency-domain signal processing techniques were proposed to mitigate FMCW and pulsed-radar interference. These techniques are capable of deleting instantaneous interference, which exists for a limited time, or bandlimited interference, which pollutes a specific portion of the whole radar band (while no solution is offered for the worst-case recurring or wideband interference [1]).

The current attitude toward interference mitigation in the industry focuses on various techniques, including pulse-to-pulse processing and removing polluted pulses, sniffing and avoiding used frequencies, using frequency diversity, using narrow main beams, or sidelobe null steering [3]. Generally, these techniques are reactive strategies that focus on eliminating interference after it occurs, making it infeasible for highly dynamic VANETs, which require for ultralow latencies. Other reactive strategies exploit the sparsity of useful signal and interference components in different transform domains, namely, the DFT and short-time Fourier transform domains, respectively, to extract the desired signal component [17] or to solve a sparse recovery problem that helps reconstruct the intervals in the range spectrum spoiled by interference [18], which belongs to a general class of time-domain excision approaches [19]. Such strategies can retain the desired target response while eliminating the interference, albeit at the cost of losing information and introducing artifacts [19].

In addition to these reactive strategies, there also exist cognitive approaches that utilize idle time-frequency resources for FMCW transmission via slope and bandwidth adaptation [20]. Interference-avoidance techniques can also be more invasive, such as notifying the driver, disabling the sensor feature, shifting functionality to another sensor, and reducing CFAR detection sensitivity [3]. However, these avoidance mechanisms either decrease radar detection performance or disable the radar completely.

Quasi-orthogonal waveforms

Concept

From the interference analysis, we established that the interference is proportional to $f = u\alpha\tau_{\max}/B$, where u is the radar duty cycle, α is the chirp slope, $\tau_{\max}$ is the maximum target round-trip time, and B is the radar bandwidth. Accordingly, by decreasing the chirp slope or, equivalently, increasing the chirp duration, interference can be mitigated. We consider a class of signals containing linear chirps with the same chirp rate, a fixed frame rate, and a fixed duty cycle. A chirp $x(t)$ from (2) and a delayed chirp $x(t-\Delta\tau)$ repeated with period T have the following power leakage/coupling for $\Delta\tau \neq 0$:

$$C(\Delta\tau) = \left| \frac{1}{T}\int_0^T e^{j[\varphi(t)-\varphi(t-\Delta\tau)]}dt \right|^2 \leq \frac{1}{\pi^2 B^2 (\Delta\tau)^2}. \tag{12}$$

Because $B\Delta\tau/T$ is the instantaneous frequency difference between the chirps, the coupling is the same as two sinusoids with frequency difference $B\Delta\tau/T$. As a result, we conclude that these waveforms are quasi-orthogonal. As $\Delta\tau$ was arbitrary, this property is maintained under random starting and arrival times of FMCW waveforms. From (12), it is possible to derive the number of signals N that cause acceptable interference, i.e., smaller than the power backscattered from a typical target. Suppose chirp duration T is divided into N segments of duration $\Delta\tau$, so that $\Delta\tau = T/N$; then, using (7),

$$P\frac{G\sigma\lambda^2}{(4\pi)^3 d^4} \geq PC(\Delta\tau)\frac{G\lambda^2}{(4\pi)^2 r^2} = P\frac{N^2}{(\pi BT)^2}\frac{G\lambda^2}{(4\pi)^2 r^2}$$
$$\Rightarrow N \leq TB\frac{\sqrt{\sigma}}{r},$$

if r and d take on similar values. For $r = 500$ m, $B = 1$ GHz, and $\sigma = 0.1\,\mathrm{m}^2$, using one long chirp of duration $T = 10$ ms leads to more than 6,000 quasi-orthogonal waveforms (compared to 12 for a conventional chirp with $T = 20\ \mu$s). The challenges of retrieving velocity and range data as well as physically realizing such a radar is briefly described in the next section.

Signal processing

In (5), even though the speed and range appear in an ambiguous combination in a single chirp, this expression is only an approximate representation of a Doppler shift because we neglected several constants and small terms, which are negligible for small T. For long chirps, these neglected values should be considered so that the range-speed ambiguity can be resolved within a single chirp. Formally,

$$\begin{aligned} y(t) &= r(t)x^*(t) \\ &= \tilde{\gamma}e^{j2\pi f_c \nu t}e^{-j2\pi\alpha\tau t}\mathrm{rect}_T(t-\tau)e^{j\varphi_0}e^{j2\pi t^2\nu\alpha} + w(t)x^*(t), \end{aligned} \tag{13}$$

where $r(t) = r_0(t)$ with $r_k(t)$ being defined in (3), $\tilde{\gamma}$ is a real quantity denoting the target reflectivity, and $\varphi_0 = \pi\alpha\tau(\tau - 2f_c/\alpha)$ is an absolute phase. The Fourier transform $Y(f)$ of $y(t)$ will have a maximum at $f^* = f_c\nu - \alpha\tau$, with

$$Y(f^*) = \tilde{\gamma}e^{j\varphi_0}\int_0^T e^{j2\pi t^2\nu\alpha}\mathrm{d}t. \tag{14}$$

Hence, finding the maximum of $|Y(f)|^2$ yields an estimate of $f_c\nu - \alpha\tau$, which allows us to write $\varphi_0 = \pi((f_c\nu - f^*)/\alpha) \times (f_c(\nu-2) - f^*)$. We can then invert (15) to solve for $\nu = 2v/c$. The solution must be found numerically, making use of the fact that the target reflectivity $\tilde{\gamma}$ is real and positive and that the complex angle of the integral increases monotonically with velocity within a significant velocity span.

When comparing the outlined approach to that of the slow-/fast-time FMCW radar case mentioned previously, note the principal difference is that instantaneous velocity is determined by measuring the complex amplitude. For pulse-Doppler radar, velocity is found by locating the target peak in the Doppler spectrum. The present approach assumes that just one target is present for any beat frequency, whereas pulse-Doppler radar may handle several targets within the same range/Doppler resolution cell. On the other hand, the attained beat-frequency resolution is refined in proportion to the prolonged sweep. In effect, both methods exhibit the same low probability that two separate targets should be superimposed within the same resolution bin.

The single slow-sweep method for obtaining nearly orthogonal waveforms contains a further challenge compared to that of pulse-Doppler radar, apart from the novel signal processing required. Indeed, the steep ramps of pulse-Doppler FMCW frequency offset the target response as compared to those of the transmit signal. Leakage between the Tx and receiver (Rx) will manifest itself at the mixer output as a high-amplitude low-frequency peak. In a typical FMCW radar, this unwanted response is removed using a configurable analog electronic filter; however, the implementation of analog high-pass filters with very low cutoff frequencies at the chip level is not feasible. Presently, radar is based on the subtraction of leakage appearing as a dc component with a feedback loop rather than filtering with respect to range and Doppler-frequency offsets. The conceived radar scheme in Figure 7 indicates that (15) can be slightly modified by digitally modifying the signal phase by some offset, thereby changing the unambiguous speed range for (15). To effectively cover a sufficiently large velocity span, two or three such velocity channels should be processed in parallel. The overall processing burden still remains fully reasonable.

Implications

The freedom of multiple orthogonal radar channels can be brought into practice in different ways. The considered case with a very large number of channels allows for the convenient method of simply not requiring any common scheduling of the channels adopted (apart from the several radars that may be located in the same vehicle, in which case channel coordination is easy to achieve). The number of vehicles in such close proximity to each other that an interference conflict is imminent will be much smaller than the number of available channels. Just by selecting the radar channels randomly, the chances are good that there will be no interference. In the case of interference, the individual vehicle will then randomly pick another channel and, with high probability, the conflict is thereby resolved.

Coordinated transmission via wireless communication

Another way to mitigate interference is by coordinating automotive radars through communications so that radars are assigned disjointed frequency-time-space resources, making use of the fact that 1) radars receive over a small fraction B_s/B of the available bandwidth, 2) radars are only active a fraction u of the time, and 3) radar signals are blocked (mostly by other vehicles) and limited by the FOV. The assignment requires coordinating the allocation of frequency-time resources through distributed network communication. Such communication can be achieved either via a dedicated technology, such as the 802.11p standard or cellular vehicle-to-everything (C-V2X) communication.

Alternatively, one can exploit the similarity of the radar circuitry to standard communication hardware and upgrade automotive radars to joint radar communication units (RCUs), which use the same hardware both for radar and communications

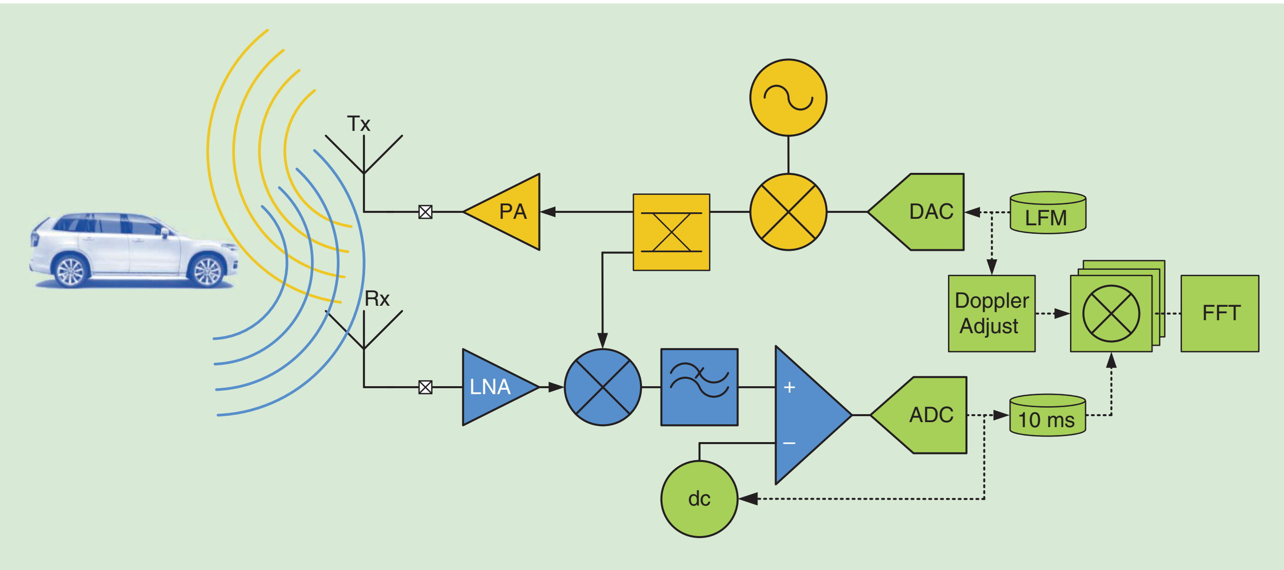

FIGURE 7. The simplified scheme of a single slow-sweep FMCW radar, which allows for many orthogonal signals. The linear FM (LFM) waveform is obtained digitally at baseband and is upconverted in the analog domain. This upconverted signal is mixed with the receive signal yielding downconversion to a signal with bandwidth essentially set by 50 MHz. The dominant power in the downconverted signal will come from the transmit signal which, however, appears as a dc component. This is removed by a dc canceler stage after which the signal is digitized. The canceler operates on the principle of minimizing at ADC output at some rate much slower than the sweep time and has the vital function of avoiding ADC saturation. In the digital domain further mixing with Doppler offsets may be required to cover a very wide range of target speeds. DAC: digital–analog converter.

(implementing data link and physical layer functions). The radar and communication functionality are time multiplexed, where communication occurs over a fixed and dedicated communication bandwidth (with the bandwidth limited by the ADC), which is free of radar transmissions. The time multiplexing is possible because of the idle period. As a result, when $u \approx 1$, RCUs cannot be used and 802.11p or C-V2X is more appropriate. Nevertheless, in this article, we consider the RCU approach, as it is readily modified when using another dedicated communication technology. As is typical in VANETs, communication is unacknowledged when using distributed MAC-based carrier-sense multiple access (CSMA).

The goal of the communication is to assign radars to time slots so that different radars remain quasi-orthogonal. In contrast to the long chirps described previously, we consider standard short chirps, thus limiting the number of transmissions per chirp period. The frequency-time resources are shared for three RCUs, namely, r_i, r_j, and r_k, as illustrated in Figure 8. The basic principle is as follows. Each vehicle initially assigns starting times to the automotive RCUs mounted on this vehicle through a central processor. These starting times are broadcast to neighboring vehicles during a communication slot. All the RCUs on a vehicle broadcast short control communication packets at the same time over the communication band. The broadcast communication packet includes information about the starting times used by all the RCUs on that vehicle. The other RCUs or vehicles that receive this information store it in a database and allocate themselves nonoverlapping starting times based on the stored information. A priority index is used to prioritize the dissemination of the resource allocation of a larger group of vehicles to avoid fluctuations in the distributed VANET.

A practical implementation requires synchronization among radars, which can be achieved through GPS or via a dedicated synchronization protocol. Using reasonable synchronization requirements (around a 1–2-μs error), the authors in [21]

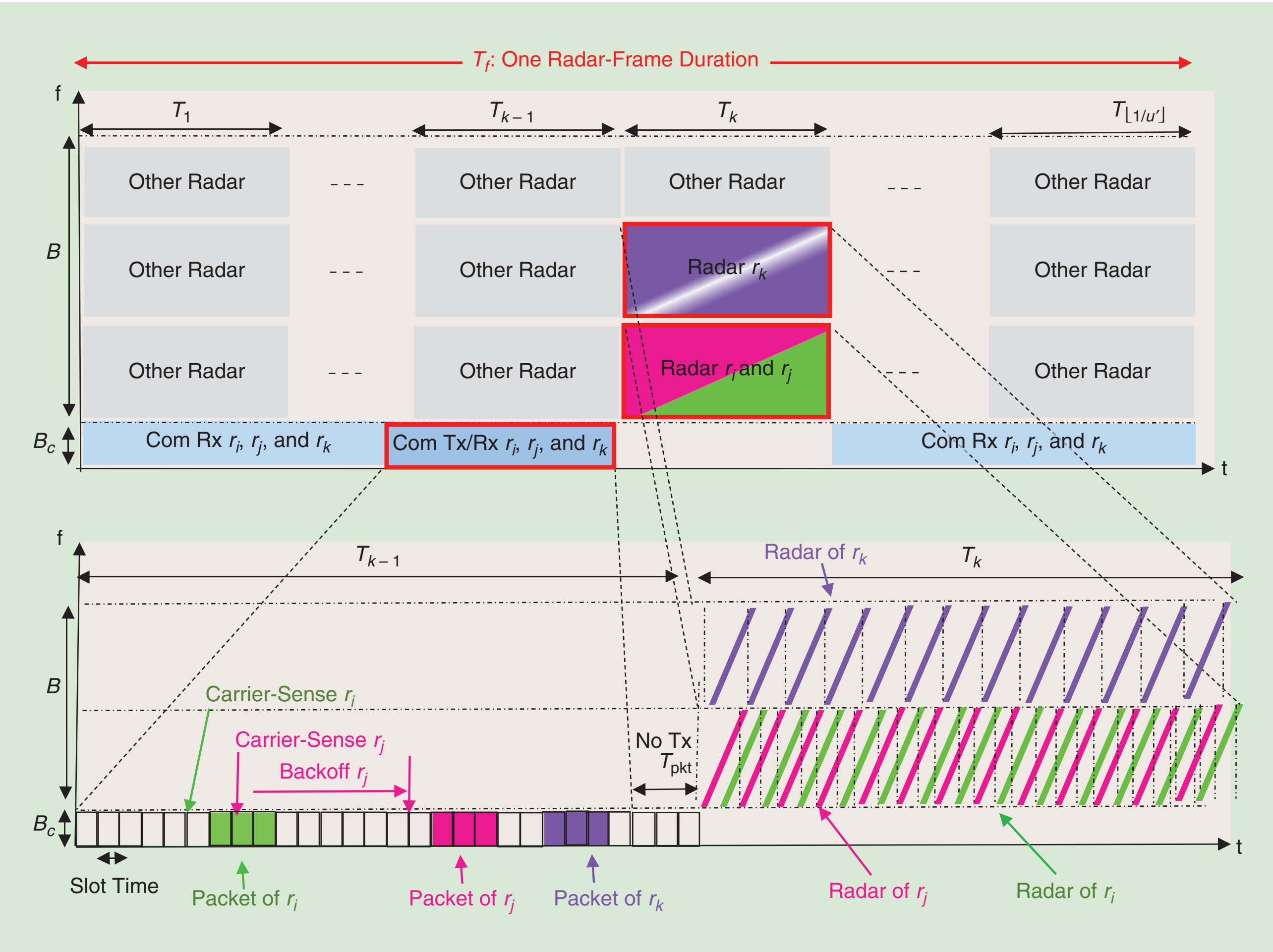

FIGURE 8. The sharing of frequency-time resources by three RCUs, r_i, r_j, and r_k. The whole bandwidth is divided into a communication channel of bandwidth B_c and sweep bandwidth B, where various types of automotive radars may use portions of B. One radar frame T_f is divided into time slots T_k, $1 \le k \le \lfloor 1/u' \rfloor$, where $u' = (N+1)T/T_f$ is the modified radar duty cycle. The radar frame is divided into time slots, where radar transmission/reception takes place in T_k, communication transmission is done prior to radar at T_{k-1}, and the reception at any time except T_k. Each RCU broadcasts its packet over the communication channel through carrier-sense multiple access, which includes the starting time and frequency band information of its radar transmission. The interference is resolved because r_i and r_j are allocated to quasi-orthogonal time slots, whereas r_k is allocated to a different frequency band. Any conflicts in frequency-time resource sharing that occur in one frame T_f are resolved in the following frames. Com: communication.

demonstrated significant reductions in radar interference within a few tens of ms. Figure 9 shows the interference probability (i.e., the expected value of f) as a function of the number of interfering radars, with and without coordination. The potential of such protocols to adapt radar signals according to changing traffic conditions via communications offers intelligent radar sensing strategies, such as cooperative localization, disabling unnecessary sensing, and so on.

Joint radar and communication

A third, more forward-looking alternative is to exploit the fact that radar and communication systems operate in similar frequencies and develop a system that can perform the dual role of radar and communication [22], called *RadCom*. Both the pilot and the data from the transmitted signal can be exploited for radar functions when processing the backscattered communication signal. A prominent candidate for this is OFDM, which is the de facto waveform for all cellular and Wi-Fi-based standards, due to its flexibility and robustness to wireless propagation effects. OFDM has also been studied extensively as a radar waveform [24]–[28] but is limited by its ADC bandwidth (generally orders of magnitude smaller than the radar bandwidth), which in turn limits radar resolution. Assuming a high-FMCW chirp slope, the OFDM and FMCW radars with identical waveform parameters exhibit similar performance in terms of accuracy and resolution [23]. OFDM provides the additional capability of communications at the expense of increased hardware complexity due to OFDM modulator/demodulator operations [23].

A way around this problem is the use of stepped-frequency OFDM, which involves consecutive OFDM frames, each transmitted with a different carrier frequency [29], [30]. The main rationale behind the use of stepped-frequency OFDM as a RadCom waveform is to surpass the range-resolution limitation of conventional OFDM radar (which is imposed by the ADC bandwidth) via frequency hopping across individual OFDM frames with low baseband bandwidth [30], while maintaining standard wideband OFDM as a special case.

Figure 10(a) illustrates the exemplary time-frequency plot of a stepped-frequency OFDM waveform. To avoid interference, different vehicles are assigned orthogonal resources, as shown for three vehicles. Hence, the stepped-frequency OFDM can exploit high resolution offered by the large total bandwidth $MN\Delta f$ using the joint processing of M individual OFDM frames on different carriers, while simultaneously requiring a low-rate ADC to sample small baseband bandwidth ($N\Delta f$) OFDM blocks. For each carrier, L OFDM symbols of duration T_{sym} are sent, constituting a frame. The choice of N, M, L, and Δf and the hopping pattern provides flexibility in the RadCom waveform and enables us to provide radar performance similar to that of an equivalent wideband OFDM radar but with low-rate, low-cost ADCs [30].

Signal processing and resource allocation

Under standard assumptions (i.e., a cyclic prefix longer than τ [26], [28] and a small Doppler approximation [24]), the received symbol on the nth subcarrier for the ℓth symbol of the mth frame [considering the same radar environment with a single target as specified in (3)] can be written as [30]

$$y_{m,\ell,n} = \gamma x_{m,\ell,n} e^{-j2\pi(f_m+n\Delta f)\tau} e^{j2\pi f_0(mL+\ell+1)T_{\text{sym}}\nu} + w_{m,\ell,n}, \quad (15)$$

where $x_{m,\ell,n}$ denotes the complex data or pilot symbol, γ is the complex channel gain, and $w_{m,\ell,n}$ is the additive noise term with variance σ^2. Delay estimation can be performed by matched filtering the data cube $y_{m,\ell,n}$ across frame-frequency dimensions (m and n directions), while processing along frame-time dimensions (m and ℓ directions) can provide a Doppler estimate. Frequency hopping across consecutive OFDM frames introduces delay-Doppler coupling, which can be overcome by incorporating phase correction terms in the DFT implementation of matched filtering [30].

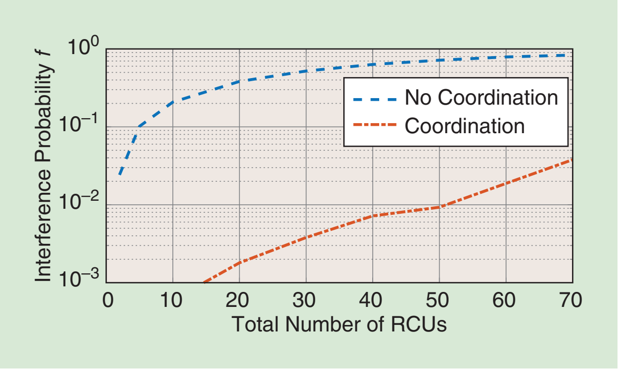

FIGURE 9. The radar interference probability f after one time frame (20 ms) with a coordinated transmission for various numbers of radars for 10,000 Monte Carlo simulations, $B = 1$ GHz, $B_s = 50$ MHz, $T = 20\ \mu$s, $f_c = 79$ GHz, and $K = 99$.

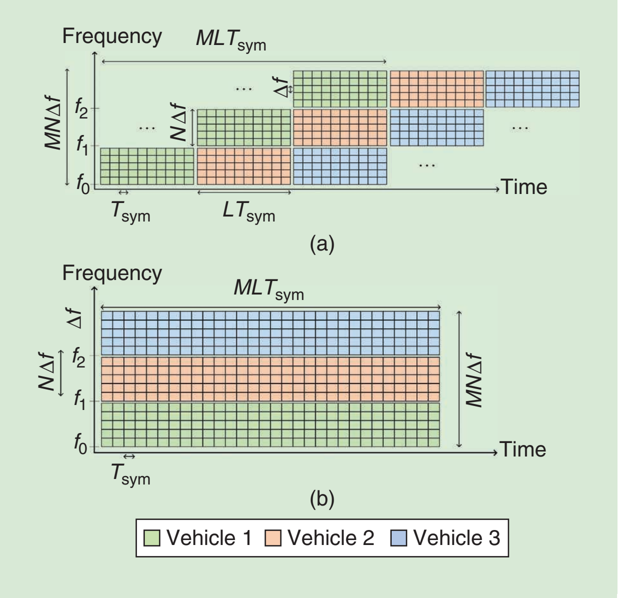

FIGURE 10. A time-frequency resource coordination in a joint radar communications vehicular network for (a) a stepped-frequency OFDM and (b) a narrow-band OFDM.

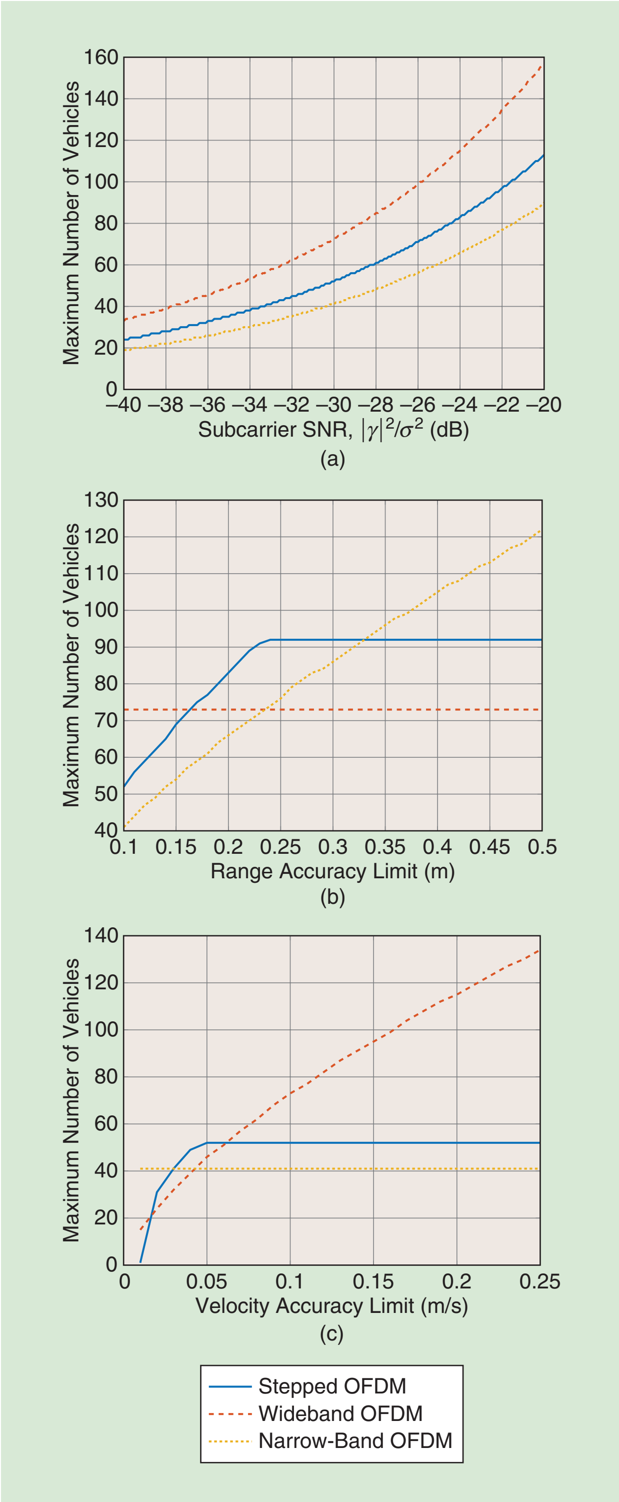

FIGURE 11. The maximum number of vehicles that can be accommodated within a time-frequency resource defined by $B_{tot} = 1$ GHz and $T_{tot} = 30$ ms for the three different OFDM waveforms, where $f_0 = 77$ GHz, $\Delta f = 500$ KHz, and $T_{cp} = 400$ ns. Stepped OFDM and narrow-band OFDM schemes require an ADC with a 50-MHz rate, whereas wideband OFDMs require a 1-GHz-rate ADC. (a) The subcarrier SNR, where range and velocity accuracy limits are 0.1 m and 0.1 m/s, respectively. (b) The range-accuracy limit, where the subcarrier SNR is –30 dB and the velocity accuracy limit is 0.1 m/s. (c) The velocity accuracy limit, where subcarrier SNR is –30 dB and the range accuracy limit is 0.1 m.

A time-frequency resource allocation scheme coordinated by a central unit (e.g., a 5G base station) helps to alleviate mutual interference among radars on different vehicles, similar to that of conventional OFDM radar networks [26, Ch. 4]. Resources can be assigned to maximize the number of vehicles that can be fit into a given time-frequency block such that each vehicle meets preset radar accuracy requirements. To characterize radar accuracy, we employ the Cramér–Rao bound [26] on variances of unbiased estimates of delay and Doppler parameters using the signal model in (15).

Figure 11(a)–(c) presents exemplary results for the three different OFDM schemes [i.e., stepped frequency, narrowband, and wideband where, in the latter case, each vehicle uses the total bandwidth (and thus requires an ADC with a 1-GHz sampling rate)] for duration LT_{sym} and then remains silent for duration $(M-1)LT_{sym}$). As shown in the figure, the stepped-frequency OFDM radar can support more vehicles in a given spectral resource than it can in the conventional narrow-band OFDM radar using the same hardware requirements because the former offers the flexibility to trade off a decrease in Doppler accuracy for an improvement in ranging accuracy (frequency hopping increases ranging accuracy and reduces Doppler accuracy). In Figure 11(b), because wideband OFDM is essentially limited by the velocity accuracy constraint, relaxing the range accuracy constraint does not further improve its performance [similar to the narrow-band OFDM in Figure 11(c)].

Finally, we note that the stepped OFDM provides a design tradeoff between the narrow-band and wideband OFDM schemes, retaining the improved resolution and accuracy properties of the wideband OFDM using significantly reduced hardware requirements, as in the case of the narrow-band OFDM.

Outlook and challenges

In this section, we consider the primary research and development challenges for the coming years. For communication-based interference mitigation strategies, the coexistence between radar and communication signals is an important challenge. For joint radar and communication signals, there is the potential of a revolution of cellular-type signals (e.g., 5G New Radio) to be reused for radar purposes [22], opening new synergies and avoiding the need for dedicated RF hardware all together. The extended frequency bands made available for 5G are interesting by themselves due to the possible improvement in radar resolution, which, together with the already standardized orthogonal signaling, establishes an exciting area for automotive sensing. The main challenge is to find solutions that will enable radar and communication functionalities with such a low information latency that vehicle safety is not compromised in any traffic scenario. These solutions should include techniques for a fair distribution of the available time and frequency space for all users. It must also secure a low data loss for both radar and communication, which, of course, is the aim of minimizing the possibility of interference.

For the generation and detection of slow chirps, new hardware architectures will be necessary; this will push a migration from analog toward digital electronics and signal processing, which will pave the way for technologies such as imaging radar. The other modulation waveforms proposed in this article will also require hardware that differs from the current radar designs. Analog-to-digital (and vice versa) conversion will be close to the RF front end, making more complex digitally generated and filtered waveforms possible. To move this development forward, we will implement a demonstrator platform, complete with mm-wave front ends, high-speed digital signal generation and acquisition, and the independent generation of arbitrary interference. The methods outlined in this article will be tested and evaluated on the demonstrator platform in a realistic environment. The intention is to use this demonstrator (see Figure 12) to verify the theoretical analysis regarding interference probability and SINR for different types of modulation, and verify the speed measurement method used for slow-ramp modulation.

Further development would include the integration of critical electronic components in CMOS technology to ensure that a complete solution is feasible to implement commercially. Advanced CMOS technologies can also facilitate the implementation of alternative waveforms on automotive radars, such as phase-modulated continuous wave (PMCW). Compared to the widely used FMCW radar, the PMCW waveform has the major disadvantage of requiring very high-rate ADCs to sample wideband code sequences. Conversely, it possesses several advantages that make it attractive for future deployments, including improved robustness to interference via proper code design, not requiring a highly linear frequency ramp synthesizer, and the inherent applicability to multiple-input, multiple-output radar configurations through code orthogonality across multiple antennas. From the perspective of radar interference mitigation and radar communications convergence, we expect that the main focus of the automotive industry in the coming years will be on the cost and integration of both analog and digital functions on the same silicon chip to reduce the likelihood of hardware failure.

Vehicle radars can also be expected to operate in higher frequency bands such as 100–300 GHz to enable more bandwidth, reduce costs, and miniaturize hardware. It is possible to influence regulators to include some level of standardization in automotive radars, which is needed for mitigating interference among different automobile brands [3], before a new frequency spectrum is made available in the higher RF bands. It is therefore timely to conduct research and discuss the developmental challenges of automotive radar interference before it becomes a problem.

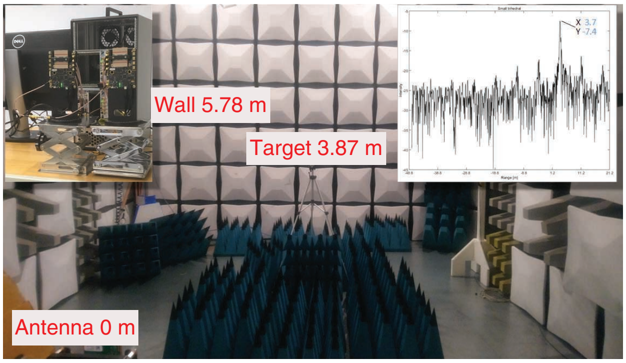

FIGURE 12. A radar test setup in an anechoic chamber with a trihedral target seen from the radar position. The upper-left inset shows the experimental radar demonstrator, while the upper-right inset shows a typical acquired range profile.

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